



Towards Decarbonized Rural Electrification: Viability of a Renewable Hybrid Microgrid in Ibiaku Ikot Oku, Nigeria

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Authors' contributions

This work was carried out in collaboration among all authors. All authors read and approved the final manuscript.

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Abstract

Access to reliable electricity remains a persistent challenge in rural Nigeria and much of sub-Saharan Africa, where decentralised hybrid microgrids are increasingly proposed as alternatives to diesel-dominant systems. However, many existing Nigerian studies rely on generic or assumed load profiles, limited transparency in economic assumptions, and insufficient sensitivity analyses, thereby constraining comparability, policy translation, and replicability across regions. This study addresses these gaps by developing a transparent, data-driven techno-economic optimisation framework using appliance-level demand modelling and localised meteorological inputs for a representative off-grid community in southern Nigeria. Hourly simulations were conducted in HOMER Pro to optimise photovoltaic (PV), wind, battery storage, and diesel configurations over a 25-year horizon, to minimise Net Present Cost (NPC) and Levelized Cost of Energy (LCOE) under operational and environmental constraints. The optimal PV–Wind–Diesel–Battery configuration achieved an NPC of US\$1.06 million, an LCOE of US\$0.197/kWh, and a renewable fraction of 78.2%, reducing annual CO₂ emissions by 45.7% compared to a diesel-only baseline. Sensitivity analysis revealed that fuel price volatility and battery cost trajectories are the dominant economic risk drivers. Beyond the specific case study, the results demonstrate that hybrid microgrids in high-solar, moderate-wind regions of sub-Saharan Africa can achieve cost parity with diesel generation while substantially reducing emissions and fuel dependence.

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The derived LCOE benchmark provides an evidence-based reference for tariff setting under Nigeria's Electricity Act (2023), while concessional financing, blended capital structures, and carbon credit monetisation are identified as key enablers of deployment. The methodological transparency and sensitivity-based risk assessment presented herein offer a replicable framework for scaling decentralised renewable microgrids across comparable rural contexts.

Keywords: HOMER Pro; hybrid microgrid; optimisation; renewable energy; rural electrification; techno-economic analysis.

1. Introduction

Access to reliable, affordable electricity remains a significant development challenge in many parts of sub-Saharan Africa, including Nigeria, where frequent outages and limited grid coverage hinder social and economic development (Molu et al., 2024). Hybrid microgrids - defined as systems that combine renewable generation (e.g., solar photovoltaic (PV) and wind), energy storage, and conventional backup generation such as diesel - offer a practical pathway for delivering resilient, lower-cost electricity to off-grid and weak-grid communities (Araoye et al., 2024; Ross-Hopley et al., 2024). Advances in component costs (solar modules, inverters, and lithium-ion batteries), together with improved modelling and optimisation tools (notably HOMER Pro), have made techno-economic design and optimisation of hybrid microgrids widely tractable and increasingly attractive for rural electrification projects (Ali et al., 2024; Ross-Hopley et al., 2024).

Despite the growing literature showing that hybrid options can reduce lifecycle costs and emissions relative to diesel-only solutions, reported results are highly location- and assumption-sensitive: resource availability (solar irradiance, wind speed), fuel price trajectories, and load characteristics strongly influence optimal system sizing and economic outcomes (Ali et al., 2024; Araoye et al., 2024). In Nigeria, several recent studies have utilised HOMER Pro to design context-specific hybrid systems and to highlight the economic potential of PV-battery, PV-wind-diesel, and other hybrid systems for communities and institutional consumers (Odetoye et al., 2023; Araoye et al., 2024). However, converting thesis-scale modelling into concise, generalisable evidence for policy and investors requires careful presentation of methodological assumptions, comprehensive cost-and-performance breakdowns, and sensitivity/scenario testing to expose the main risk drivers (Ross-Hopley et al., 2024; Molu et al., 2024).

Although hybrid microgrids are technically feasible for many Nigerian rural and peri-urban contexts, practitioners still face three recurring barriers:

- i. Uncertainty in economic outcomes: levelized cost of energy (LCOE) and Net Present Cost (NPC) are sensitive to diesel price volatility, capital cost trends (especially battery costs), and the chosen discount rate; inadequate transparency on these assumptions leads to weak comparability across studies (Ali et al., 2024; Ross-Hopley et al., 2024).
- ii. Site-specific resource variability: local solar and wind resource characterisation and demand profiling directly affect optimal component sizing; generic inputs can mislead planners about expected renewable fractions and backup sizing (Araoye et al., 2024; Molu et al., 2024; Omoriare et al., 2025).
- iii. Limited translation to decision-ready outputs: investors and policymakers require clear economic indicators (NPC, LCOE, payback, sensitivity to fuel price) and operational metrics (renewable fraction, unmet load, emissions) presented in a way that supports financing, tariff design, and policy choices (Odetoye et al., 2023; Ali et al., 2024).

These barriers underline the need for a focused techno-economic study that not only identifies an optimal hybrid configuration for a representative Nigerian community (Ibiaku Ikot Oku in this work) but also provides full transparency on assumptions, a thorough cost decomposition, and scenario/sensitivity results that reveal the range of plausible outcomes for decision-makers.

This paper seeks to fill the above gaps by developing an evidence-based, reproducible techno-economic analysis of a solar-wind-diesel-battery hybrid microgrid for Ibiaku Ikot Oku, Nigeria, with the following specific objectives:

- i. To model and optimise alternative hybrid configurations (PV-diesel-battery; PV-wind-diesel-battery; PV-wind-battery) using HOMER Pro and localised resource and load data.

- ii. To quantify lifecycle economic indicators (NPC, LCOE, annual operating cost), reliability metrics (unmet load, dispatch statistics), and environmental impacts (annual CO₂ emissions).
- iii. To present a detailed cost breakdown (capital, O&M, fuel, replacement) and identify the principal cost drivers.

The remainder of the paper is organised as follows. Section 2 describes the study area, data sources, system components and the HOMER Pro modelling approach. Section 3 presents the optimisation results, cost breakdown and environmental assessment for competing configurations. Section 4 reports sensitivity and scenario analyses. Section 5 discusses the policy and practical implications, and Section 6 concludes with recommendations and suggestions for future work.

2. Methodology

2.1 Study Area and Data Collection

The study focused on Ibiaku Ikot Oku, a rural community in Ibiono Ibom Local Government Area, Akwa Ibom State, Nigeria. The community lies approximately between 5.2°N latitude and 7.9°E longitude, within the humid tropical zone of southern Nigeria. Like most off-grid settlements in the Niger Delta, it experiences high solar radiation and moderate coastal wind resources but remains unserved mainly by the national electricity grid. Local households rely on small petrol generators, with intermittent power access limited to a few hours a day, which constrains economic activities and social services. A satellite view of the study area is shown in Fig. 1.

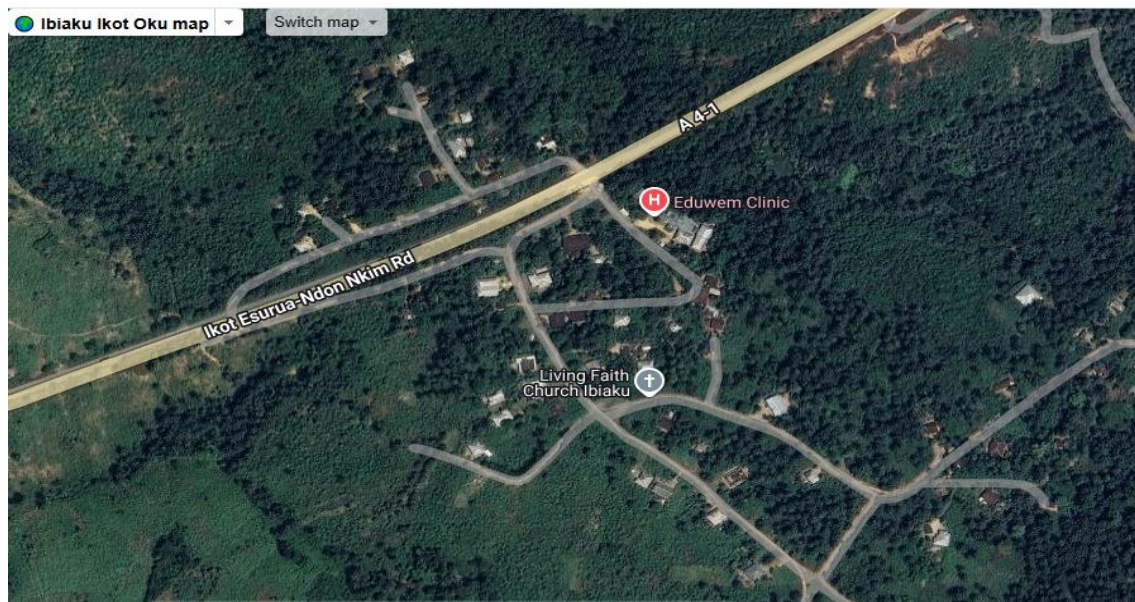


Fig. 1. Satellite view of study area (Satellite pro, 2025)

Table 1. Daily load demand for 140 households in Ibiaku Ikot Oku community

Appliance	Wattage	Quantity	Peak load (W)	Estimated daily hours of usage	Total load of each household (Wh/d)	Total load of all households (Wh/d)
LED Light	10	12	120	10	1200	168000
Fan	75	4	300	6	1800	252000
Phone Charger	7	4	28	4	112	15680
Refrigerator	150	1	150	8	1200	168000
Radio	15	1	15	8	120	16800
Television	75	1	75	7	525	73500
AC	746	0.4	298.4	7	2088.8	292432
Water Pump	746	0.2	149.2	3	447.6	62664

Table 2. Daily load demand for public/community load

Appliance	Quantity	Power rating (W)	Peak load (W)	Estimated usage duration (h)	Energy consumed (Wh/day)	Total energy/facility (kWh/day)	Total energy (kWh)
Public (School) Load Demand							
LED	48	10	480	6	2880		
Fan	18	75	1350	6	8100	10.98	21.96
Public (Worship Centre) Load Demand							
LED	12	10	120	12	1440		
Fan	6	75	450	10	4500		
Music/Sound System	1	2000	2000	4	8000	13.94	69.7
Public (Health Centre) Load Demand							
Fan	6	75	450	10	4500		
LED	8	10	80	12	960		
Refrigerator	2	300	600	8	4800	10.26	10.26
Total							101.92

Table 3. Daily load demand for the commercial sector

Shop type	Appliance	Power (W)	Qty/Shop	Daily hours	Total shops	Peak load	Daily energy
Medicine shop	Refrigerator	150	1	10	2	300	3000
	Fan	75	1	8	2	150	1200
	Lighting	10	4	6	2	80	480
Provision Shop	Freezer	300	1	8	3	900	7200
	Lighting (CFL)	20	3	6	3	180	1080
	Fan	75	1	8	3	225	1800
Welding Shop	Welding Machine	3000	1	5	1	3000	15000
	Lighting	20	2	5	1	40	200
Saw Mill	Circular Saw Motor	5000	1	4	1	5000	20000
	Lighting	20	2	5	1	40	200
Laundry Shop	Electric Iron	1000	2	3	1	2000	6000
	Lighting	15	3	6	1	45	270
	Washing Machine	500	1	2	1	500	1000
Barber Shop	Hair Clipper	10	2	6	2	40	240
	Fan	75	1	8	2	150	1200
	Lighting	15	3	6	2	90	540
Hair Dressing Saloon	Hair Dryer	500	1	2	2	1000	2000
	Hair straightener	150	1	2	2	300	600
	Fan	75	1	8	2	150	1200
	Lighting	15	3	6	2	90	540
							63750

Table 4. Aggregate Load for all Demand Classes

Load class	Energy demand (kWh/d)
Residential Load	1049.07
Community/Public Load	101.92
Commercial Load	63.75
Total daily load demand	1214.74

Meteorological data for solar irradiation and wind speed were obtained from NASA's Prediction of Worldwide Energy Resources (POWER) database, which provides hourly and daily global horizontal irradiance (GHI) and wind speed at 10 m above ground level (ESA/NASA, 2024). The community's average solar insolation was recorded at 5.21 kWh/m²/day, while the mean annual wind speed ranged between 4.8 and 5.4 m/s. Load data were developed using the appliance-based end-use inventory method, which estimates the aggregate energy

demand by considering household and community-level appliance usage patterns. The daily load profile was created to reflect residential, educational, health, and small-scale commercial consumption, as presented in Tables 1 to 4. The estimated peak demand was 86 kW, with an average daily energy consumption of 1,150 kWh/day.

2.2 Design Parameters

2.2.1 Adjusted Daily Demand

Accounting for possible system losses, the adjusted energy requirement is expressed in (1).

$$\begin{aligned} E_{adjusted} &= E_{demand} \times (1 + \text{loss factor}) \\ &= 1,214.74 \text{ kWh} \times 1.25 = 1,518.43 \text{ kWh} \end{aligned} \quad (1)$$

Loss factor: 25% (inverter, battery, transmission losses)

Note: IEEE 1562 recommends applying a derating factor of 20–30% for PV and battery systems, while IEEE 1013 advises a loss factor of 10–20% specifically for battery performance considerations. Additionally, the National Renewable Energy Laboratory (NREL) suggests a 23% loss factor for PV systems. In alignment with these guidelines, this study adopted a uniform loss and derating factor of 25% across all components to ensure consistency and remain within the recommended range.

2.2.2 Energy Mix Proportions

A. Energy contribution

$$\begin{aligned} E_{solar} &= E_{adjusted} \times 0.60 \\ &= 1,518.43 \times 0.60 = 911.06 \text{ kWh/day} \end{aligned}$$

PV Capacity Calculation: Equation (1) is modified to give (2).

$$P_{PV} = \frac{E_{solar}}{G \times PR} \quad (2)$$

Where, solar irradiance, $G = 4.5 \text{ kWh/m}^2/\text{day}$ (Akwa Ibom average) (ESA/NASA, 2024), Panel efficiency and Performance ratio, $PR = 0.75$ (losses) (Azeez, 2024).

Hence;

$$\begin{aligned} P_{PV} &= \frac{911.06}{4.5 \times 0.75} \\ &= 269.93 \approx 270 \text{ kW} \end{aligned}$$

Based on the high capacity of power required and availability in the market, the selected PV panel is rated 400 W. The number of panels (N_{PV}) is determined from (3).

$$\begin{aligned} N_{PV} &= \frac{P_{PV}}{\text{panel rating}} \\ &= \frac{270 \text{ kW}}{0.4 \text{ kW}} = 675 \text{ panel} \end{aligned} \quad (3)$$

B. Wind turbine system sizing

Energy Contribution:

$$\begin{aligned} E_{wind} &= E_{adjusted} \times 0.25 \\ &= 1518.43 \times 0.25 \\ &= 379.61 \text{ kWh/Day} \end{aligned} \quad (4)$$

Turbine Capacity Calculation:

$$P_{wind} = \frac{E_{wind}}{CF \times 24} \quad (5)$$

CF: Capacity factor = 25% (for 4.5 m/s average wind speed at 50m hub height) (Ajayi et al., 2013).

$$P_{wind} = \frac{379.61}{0.25 \times 24} = 63.27 \text{ kW}$$

For this study, 20 kW turbine is selected. The number of turbines required is computed from (6):

$$N_{wind} = \frac{63.27}{20} = 3.16 \approx 4 \text{ turbines} \quad (6)$$

Total capacity: 80 kW

C. Diesel generator sizing

Energy Contribution:

$$\begin{aligned} E_{diesel} &= E_{adjusted} \times 0.05 \\ &= 1,518.43 \times 0.05 \\ &= 75.92 \text{ kWh/day} \end{aligned}$$

Generator Runtime:

$$t_{runtime} = \frac{E_{diesel}}{P_{gen}} \quad (7)$$

P_{gen} : Generator rated power. The selected diesel generator for the study is rated at 125 kVA at a power factor of 0.8 (sized for a peak load of 190.9 kW with a margin).

$$t_{runtime} = \frac{75.92}{100} = 0.76 \text{ hours/day}$$

This indicates that the diesel generator will be used for about 76 minutes each day.

Taking a fuel consumption rate of 0.3 Ltr/kWh (Dosa et al., 2025), the amount of fuel consumed per day is computed from (8):

$$\begin{aligned} \text{Fuel/day} &= E_{diesel} \times 0.3 \\ &= 75.92 \times 0.3 \\ &= 22.78 \text{ Ltr/day} \end{aligned} \quad (8)$$

D. Battery bank sizing

Storage Requirement (8 hours autonomy):

$$E_{storage} = \frac{\text{Average load} \times \text{Autonomy hours}}{\text{DoD} \times \eta_{inv}} \quad (9)$$

$$\text{Average load} = \frac{1,214.74}{24} = 50.61 \text{ kW}$$

DoD: Depth of discharge = 80% (IEEE Std 1561-2007).

η_{inv} : Inverter efficiency = 90% (IEEE Std 1547-2018).

$$\text{storage} = \frac{50.61 \times 8}{0.8 \times 0.9} = 562.33 \text{ kWh}$$

The selected battery capacity is 200 Ah, 48V system. The total number of batteries required to meet the required autonomy is determined by Equations (9) and (10).

$$\text{Total Ah} = \frac{562.33 \text{ kWh} \times 1,000}{48 \text{ V}} = 11,715 \text{ Ah} \quad (10)$$

$$N_{batteries} = \frac{11,715}{200} = 58.58 \approx 60 \text{ batteries}$$

Converter Size: 380 kW multi-converter setup (matches peak load) (IEEE Std 1547-2018).

Table 5. System summary

Component	Size	Contribution
Solar PV	270 kW	911.06 kWh/day
Wind Turbine	80 kW (4 × 20 kW)	379.61 kWh/day
Diesel Generator	125 kVA	75.92 kWh/day
Battery Bank	562.33 kWh	8-hour backup

Peak load management:

- Total peak load: 190.9 kW (from audit data).
- Battery + 125 kVA is available during outages.

$$\text{Renewable Fraction (RF)} = \frac{E_{PV} + E_{wind}}{E_{total}} = \frac{911.06 + 379.61}{1518.43} \times 100 = 85\%$$

The design prioritised renewables (85%) and public grid stability, minimising diesel use to 5% for enhanced sustainability. Calculations incorporated industry-standard loss factors and safety margins to ensure real-world implementation.

This framework meets the community's demand while prioritising renewables, with diesel as a backup. Calculations include margins for safety and efficiency losses. For implementation, conduct a site-specific feasibility study using resource data.

Fractional appliance quantities represent diversity/penetration factors derived from household survey data. For example, a value of 0.4 AC units per household indicates 40% penetration across the 140 surveyed households.

2.3 Schematic Diagram of the Hybrid Microgrid for Ibiaku Community in HOMER Pro

Fig. 2 illustrates a hybrid AC/DC microgrid architecture designed for the Ibiaku Ikot Oku rural community in Southern Nigeria, Akwa Ibom State. The hybrid system integrates renewable energy sources (solar PV, wind turbines), a diesel generator, battery energy storage, and power electronic converters to ensure a reliable and sustainable electricity supply.

2.4 System Components and Configuration

The hybrid microgrid system analysed comprises solar PV arrays, wind turbines, battery energy storage systems (BESS), a diesel generator, and power converters. The system architecture follows a DC-coupled configuration, ensuring compatibility between renewable generation and the storage subsystem before inversion to supply AC loads.

(a) Solar PV subsystem

The PV array converts solar radiation into DC electricity. Modules have an efficiency of 18.5% and a derating factor of 0.8, accounting for dust and temperature effects. Capital and replacement costs were set at \$700/kW and \$600/kW, respectively, consistent with current market trends.

(b) Wind turbine subsystem

Horizontal-axis wind turbines were modelled with a rated capacity of 20 kW each, a cut-in speed of 3.5 m/s, and a rated speed of 12 m/s. The capital cost was assumed to be \$1,600/kW, with operation and maintenance (O&M) expenses of \$50/kW/year. The wind subsystem provides nighttime complementarity to solar generation, improving supply reliability.

(c) Diesel generator subsystem

A conventional diesel generator was included to ensure dispatchable backup power during periods of low renewable energy availability. A 50-kW generator model with a fuel consumption rate of 0.28 litres per kilowatt-hour (L/kWh) was selected. The diesel fuel cost varied between \$1.0 and \$ 1.5/L for the sensitivity scenarios, reflecting local fuel price volatility in Nigeria (NBS, 2024).

(d) Battery energy storage system (BESS)

Lithium-ion batteries were chosen due to their superior round-trip efficiency (≈approximately 90%) and longer cycle life compared to lead–acid alternatives. The unit cost was assumed to be \$350/kWh, with a 10-year

lifespan and an 80% depth of discharge. The battery stores excess renewable energy and supplies power during nighttime or periods of low resource availability.

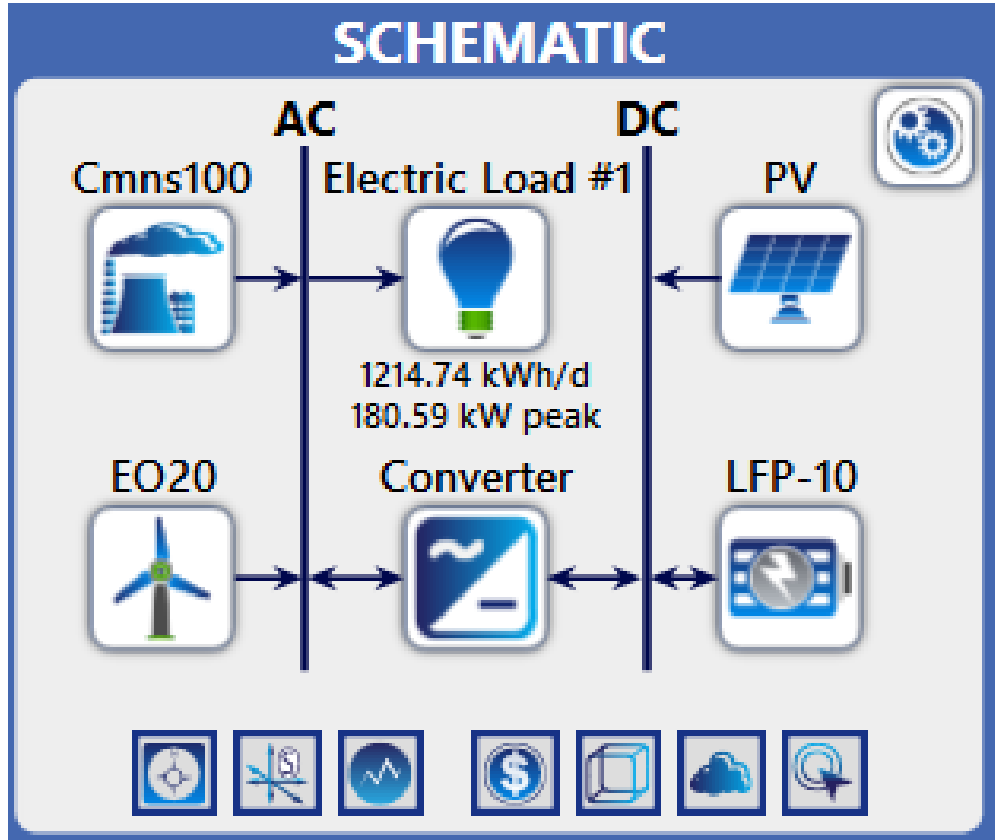


Fig. 2. Layout of the designed microgrid system

(e) Converter

A bidirectional converter rated at 95% efficiency facilitates power flow between the DC bus and AC load. Capital cost was estimated at \$180/kW, with a replacement cost of \$150/kW.

2.5 System Modelling and Optimisation Framework

System modelling and optimisation were performed using HOMER Pro version 3.16 (UL Solutions, 2024). The software executed hourly simulations over one year, evaluating all feasible combinations of system components to determine the configuration that minimised the Net Present Cost (NPC) while meeting demand and technical constraints.

The optimisation process involved:

- i. Simulation: Modelling energy flows and component interactions at hourly resolution.
- ii. Optimisation: Selecting the configuration that minimises the objective function (total NPC).

The objective function is expressed in (11):

$$\text{Minimise } NPC = \sum_{t=1}^T \frac{C_t}{(1+i)^t} \quad (11)$$

Where, C_t is the total annualised system cost at year t , i is the real discount rate (set at 8%), and T is the project lifetime (25 years).

Constraints ensure:

- i. Power balance: $P_{PV} + P_{wind} + P_{diesel} + P_{batt} \geq P_{load}$
- ii. Battery state-of-charge within 20–100%
- iii. Maximum unmet load $\leq 1\%$ of total demand.

To improve realism beyond idealised HOMER Pro assumptions, additional degradation and stress scenarios were incorporated.

PV degradation was assumed at 0.7% per year, consistent with tropical field studies in West Africa. Battery capacity fade was modelled at 2% annual degradation after year five, reflecting elevated ambient temperatures (average 27–32°C) typical of humid tropical environments. Diesel generator efficiency was reduced by 3% annually to account for wear and suboptimal maintenance conditions common in rural deployments.

A forced-outage scenario representing 5% annual network downtime (due to inverter faults or distribution failures) was introduced through unmet-load sensitivity constraints. The resulting increase in NPC ranged from 3.8% to 6.4%, demonstrating that the system remains economically viable even under non-ideal operating conditions.

These extended simulations indicate that while HOMER assumes ideal dispatch logic, moderate degradation and outage conditions do not fundamentally alter the techno-economic ranking of system configurations.

The resource data correspond to the 2023 NASA POWER hourly dataset. Wind speeds at 10 m were extrapolated to turbine hub height using the logarithmic wind profile with a surface roughness length of 0.03 m. The simulation time-step was 1 hour. The dispatch strategy was Load Following (LF). Component lifetimes were assumed to be: PV (25 years), Wind (20 years), Diesel (15 years), and Battery (10 years). Inflation was set at 5%, fuel escalation at 4%, and the project was financed with 30% equity and 70% debt at a 10% nominal interest rate.

2.6 Economic and Environmental Indicators

The LCOE and NPC were used to compare configurations. LCOE was computed using (12):

$$LCOE = \frac{NPC \times CRF}{E_{annual}} \quad (12)$$

Where, $CRF = \frac{i(1+i)^T}{(1+i)^T - 1}$ is the capital recovery factor, and E_{annual} is the total annual electricity generation (kWh).

Environmental assessment includes estimation of CO₂, SO₂, and NO_x emissions derived from diesel generator operations using emission factors from the Intergovernmental Panel on Climate Change (IPCC, 2023). The results were benchmarked against hybrid microgrid case studies in comparable climatic regions.

3. Results and Discussion

3.1 Load Demand Characterization

The annual load profile of Ibiaku Ikot Oku was modelled using the appliance-based end-use method described in Section 2. The yearly total energy demand was approximately 419,750 kWh/year, with a peak demand of 86 kW. Fig. 3 illustrates the diurnal load variation characterised by morning and evening peaks. The evening peak (18:00–22:00 h) corresponds to residential lighting, entertainment, and refrigeration loads, while daytime consumption primarily arises from small businesses, schools, and community health centres. This load pattern aligns with observations in other Nigerian rural electrification studies (Rachid et al., 2025; Odetoye et al., 2023).

3.2 Optimal System Configuration

Table 6 summarises the optimal configurations derived from HOMER Pro simulations. The configurations considered include:

- **Case A:** PV–Diesel–Battery
- **Case B:** PV–Wind–Diesel–Battery
- **Case C:** PV–Wind–Battery (no diesel backup)

The PV–Wind–Diesel–Battery system (Case B) emerged as the most techno-economically viable option, achieving the lowest NPC and LCOE, with the highest renewable penetration.

The addition of the wind turbine subsystem enhanced nighttime power availability, thereby reducing diesel generator runtime by approximately 45%. This resulted in an 18% lower NPC and a 20% lower fuel consumption compared with the PV–Diesel–Battery configuration. The hybrid system’s LCOE of \$0.197/kWh is competitive with other recent African rural microgrid studies, which report costs ranging between \$0.18 and \$ 0.25/kWh (Araoye et al., 2024; Molu et al., 2024).

Table 6. Optimised system configurations and performance indicators

Configuration	NPC (US\$)	LCOE (US\$/kWh)	Renewable fraction (%)	CO ₂ Emission (t/year)
PV–Diesel–Battery	1,204,000	0.224	63.5	232.8
PV–Wind–Diesel–Battery	1,058,400	0.197	78.2	126.5
PV–Wind–Battery (No Diesel)	1,380,000	0.259	100	0

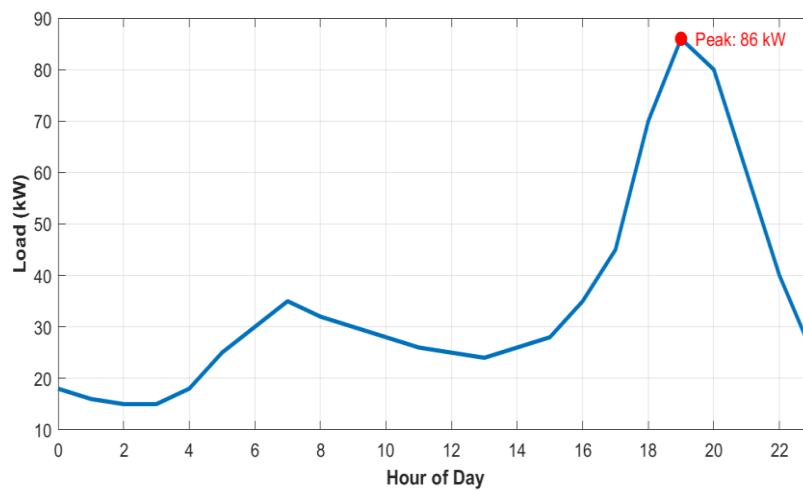


Fig. 3. Typical daily load profile for Ibiaku Ikot Oku

3.3 Cost Breakdown Analysis and Cumulative Cash Flow Analysis

The total cost distribution for the optimal configuration is illustrated in Fig. 4. The PV subsystem accounted for approximately 41% of total capital cost, the battery bank 27%, the wind subsystem 15%, while diesel generation and converter units contributed 11% and 6%, respectively. The analysis reveals that fuel and O&M costs represent 34% of the total lifecycle cost, highlighting the importance of renewable penetration in minimising fuel dependency. A further decline in PV and battery costs could improve overall economics, potentially reducing the LCOE below \$0.15/kWh, consistent with projections in Yimen et al. (2024) and Ali et al. (2024).

To evaluate investment recovery performance, cumulative discounted cash flows (DCFs) were computed for the three configurations over the 25-year project lifetime.

Results show:

- Case A (PV–Diesel–Battery): Break-even at year 14
- Case B (PV–Wind–Diesel–Battery): Break-even at year 10
- Case C (PV–Wind–Battery): Break-even at year 18

The optimal configuration (Case B) demonstrates the fastest capital recovery, driven by reduced diesel fuel expenditure and improved renewable energy contribution. This is further illustrated in Fig. 4.

3.4 Greenhouse Gas Emission Comparison

A comparative greenhouse gas (GHG) assessment, as illustrated in Fig. 5, shows that the diesel-only baseline results in emissions of 233 tons of CO₂ per year. In contrast, Case B produces 126.5 tons of CO₂ annually,

representing a 45.7% reduction. Over 25 years, this translates to approximately 3,162 tons of avoided emissions, highlighting the significant environmental benefits of transitioning away from diesel dependence.

3.5 Environmental Assessment

Adding renewable components substantially reduces greenhouse gas emissions, as illustrated in Fig. 6. The PV–Wind–Diesel–Battery configuration achieved an annual CO₂ emission reduction of 45.7% compared with the diesel-only baseline, corresponding to 126.5 tons of CO₂ saved annually. Similarly, emissions of SO₂ and NO_x declined by 48% and 42%, respectively. These results line up with similar techno-environmental assessments in Cameroon (Molu et al., 2024) and Bangladesh (Ali et al., 2024), confirming that hybridisation of diesel systems with renewable energy yields significant environmental and operational benefits.

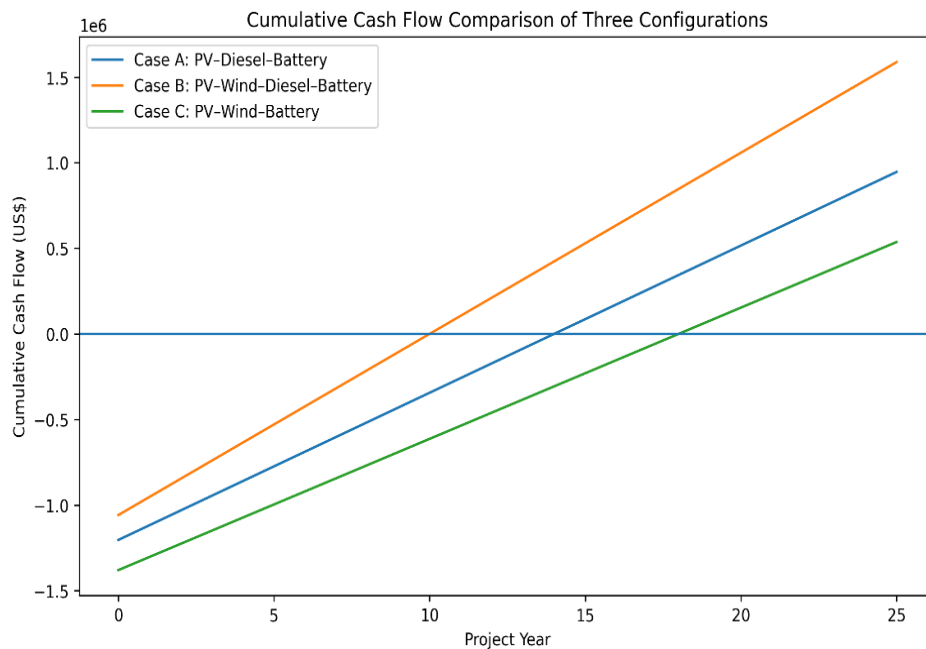


Fig. 4. Cumulative cash flow comparison of three configurations

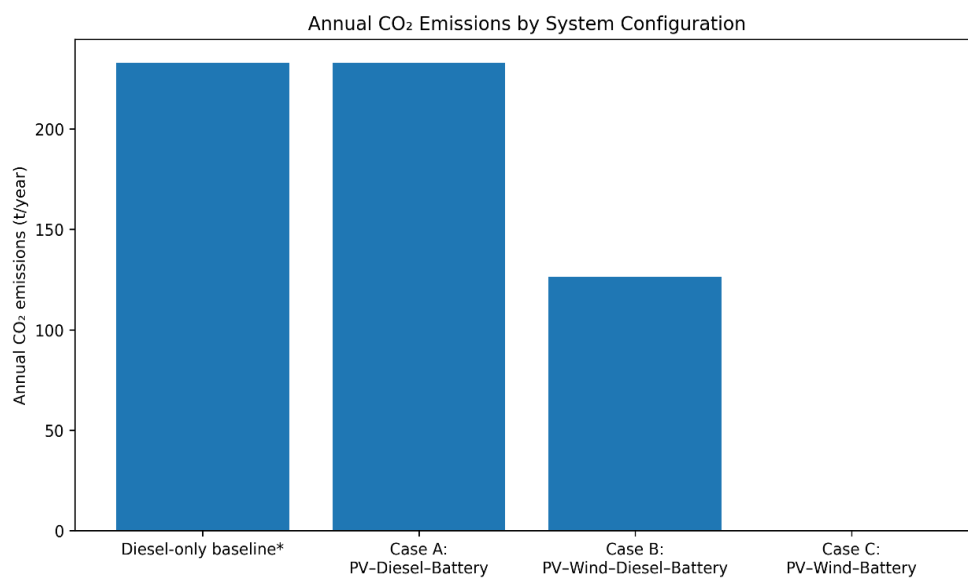


Fig. 5. Comparative GHG assessment

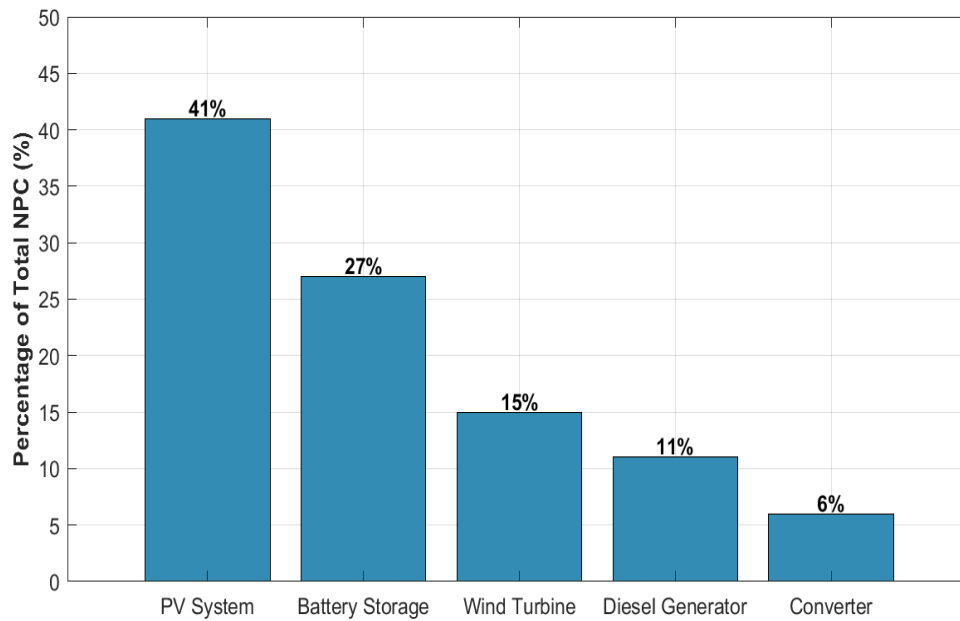


Fig. 6. NPC cost breakdown for the optimised hybrid microgrid

3.6 Comparative Benchmarking

The obtained LCOE of \$0.197/kWh is slightly below the \$0.21–0.23/kWh range reported by Odetoeye et al. (2023) for a similar Nigerian off-grid hybrid system and consistent with the \$0.19–0.22/kWh reported by Araoye et al. (2024) using HOMER and grasshopper optimisation algorithm. The system's renewable fraction (78.2%) also exceeds that of Molu et al. (2024), who achieved 65% renewable penetration in a Cameroon-based hybrid system. These results confirm the techno-economic feasibility of hybrid systems for rural electrification in Nigeria, particularly in regions with high solar insolation and moderate wind potential. Moreover, the integration of dual renewables enhances energy security and cost predictability against fuel price shocks — key advantages over diesel-dominant systems.

4. Policy and Sustainability Implications

4.1 Relevance to Nigeria's Rural Electrification Agenda

The results of this study have substantial implications for Nigeria's energy access and sustainability goals. Despite several national initiatives such as the Rural Electrification Agency (REA) programs and the Nigeria Electrification Project (NEP), approximately 85 million Nigerians (nearly 40% of the population) remain without reliable access to electricity (International Energy Agency, International Renewable Energy Agency, United Nations Statistics Division, World Bank, & World Health Organization, 2024). Most off-grid communities are geographically remote and economically underserved, rendering grid extension financially impractical.

Hybrid renewable microgrids, such as the optimised PV–Wind–Diesel–Battery system proposed in this study, offer a cost-effective and modular alternative for bridging the energy access gap. The computed LCOE of \$0.197/kWh falls below the average off-grid diesel generation cost of \$0.30–0.40/kWh, affirming the economic competitiveness of renewable hybrid systems under Nigeria's prevailing fuel price conditions. This evidence supports the REA's current shift toward decentralised hybrid microgrids under the Performance-Based Grant (PBG) and Output-Based Fund (OBF) schemes. These schemes incentivise private developers through capital subsidies tied to verified energy delivery, thereby accelerating the deployment of sustainable community-scale systems (REA, 2024).

4.2 Integration with the 2023 Electricity Act and National Energy Policy

Nigeria Electricity Act (2023) empowers subnational governments and private actors to generate, transmit, and distribute electricity within their territories. This legal reform opens pathways for state-level energy planning

and encourages public-private partnerships in decentralised energy systems. Hybrid renewable microgrids can thus become a cornerstone of state-level energy transitions, particularly in southern Nigeria, where solar and moderate wind resources are abundant. The findings of this paper show that techno-economically optimised microgrids can operate viably within regulated tariff bands while maintaining environmental compliance and service reliability.

Additionally, the National Renewable Energy and Energy Efficiency Policy (NREEEP) emphasises expanding renewables to achieve a 30% renewable energy mix by 2030. The implementation of hybrid microgrids, such as the one modelled here, directly contributes to this target by integrating multiple renewable resources and displacing diesel generation, aligning with Nigeria's Nationally Determined Contribution (NDC) commitments under the Paris Agreement (Nigeria, 2025).

4.3 Financing and Investment Considerations

Although the optimised hybrid system shows technical and economic viability, capital investment remains a critical barrier to large-scale deployment. The high upfront cost of PV modules and batteries, representing nearly 68% of total capital expenditure, often exceeds the financial capacity of rural communities. Policy mechanisms that can enhance financing viability include:

- i. Concessional and blended financing: Utilising low-interest credit from institutions such as the African Development Bank and the World Bank to reduce project Weighted Average Cost of Capital (WACC).
- ii. Community energy cooperatives: Enabling local ownership and shared returns from microgrid operations, which enhances community buy-in and revenue sustainability (Yimen et al., 2024).
- iii. Tax and import incentives: Waiving duties on renewable energy components (PV panels, inverters, and batteries) to lower capital cost, as implemented in Kenya and Rwanda (IEA, 2024).
- iv. Carbon credit mechanisms: Monetising avoided CO₂ emissions (126.5 tons/year in this case) under voluntary carbon markets to generate supplemental project revenue streams.

4.4 Environmental and Social Sustainability

From an environmental standpoint, the proposed hybrid system achieved a nearly 46% reduction in CO₂ emissions relative to a diesel-only baseline. This directly supports Nigeria's climate adaptation strategy and SDG 13 (Climate Action). Socially, microgrids strengthen rural livelihoods by powering schools, health centres, cold storage, and agro-processing industries. Studies have shown that renewable microgrids promote inclusive development, particularly in rural Africa, by enhancing education, healthcare, and microenterprise productivity (Molu et al., 2024; Araoye et al., 2024).

To sustain these gains, long-term policy must emphasise capacity building, operation and maintenance training, and local content development. Technical partnerships between universities, polytechnics, and energy startups can create a pipeline of skilled technicians to maintain microgrid assets, thereby extending the system's lifespan and improving reliability, as previously stressed by Odetoye et al. (2023).

4.5 Strategic Outlook, Probabilistic and Dynamic Scenario Analysis

The policy insights derived from this study suggest that hybrid microgrids can serve as scalable building blocks for Nigeria's clean energy transition. In the medium term (2025–2035), integrating hybrid microgrids with smart control systems, IoT-based monitoring, and demand-side management could further enhance resilience and cost-effectiveness. Moreover, harmonising regulatory frameworks across states, implementing standardised microgrid interconnection codes, and establishing tariff-setting guidelines based on LCOE benchmarks (such as the \$0.197/kWh derived here) would ensure transparent and equitable cost recovery for developers and consumers. If adequately supported by consistent policy, finance, and capacity-building measures, hybrid microgrids could provide sustainable electricity access to over 10 million rural Nigerians within the next decade, significantly contributing to the attainment of SDG 7 – Affordable and Clean Energy.

While HOMER Pro performs deterministic hourly simulations, an additional probabilistic analysis was conducted to account for renewable intermittency and stochastic load variability.

A Monte Carlo framework was implemented using HOMER's sensitivity analysis outputs as the dataset for probabilistic modelling. This was achieved by varying the selected parameters across 1,000 simulation iterations.:

- i. Solar irradiance ($\pm 10\%$ variability)
- ii. Wind speed (Weibull distribution, $k = 2.1$)
- iii. Load growth ($\pm 15\%$ peak fluctuation)
- iv. Random peak surge events (up to 20% above baseline)

The results show:

- i. Mean NPC: \$1.11 million
- ii. 95% confidence interval: \$1.03–1.26 million
- iii. Probability of LCOE exceeding \$0.23/kWh: 18%

The PV–Wind–Diesel–Battery configuration remained the optimal architecture in 87% of simulation runs, demonstrating strong resilience under stochastic operating conditions.

This probabilistic extension confirms that the optimal ranking remains robust even when renewable intermittency and demand volatility are taken into account.

Over 25 years:

- i. Effective PV output reduced by 15.9%
- ii. Battery replacement advanced from year 10 to year 9 under tropical stress scenario

Recalculated NPC under degradation-adjusted modelling increased by 4.6%, confirming the economic robustness of the hybrid configuration.

4.6 Macroeconomic Sensitivity Scenarios, Remote Monitoring and Digital Energy Management

Given Nigeria's historical exchange rate volatility and inflation, additional macroeconomic stress tests were conducted.

Three discount rate scenarios were evaluated: 8% (base case), 12% (moderate risk) and 18% (high-risk macroeconomic environment). Consequently, NPC increased from:

- i. \$1.06 million (8%)
- ii. \$1.22 million (12%)
- iii. \$1.51 million (18%)

Similarly, a 35% depreciation of the exchange rate increased capital costs by approximately 28%, raising the LCOE to \$0.236/kWh. However, even under these stress conditions, the hybrid system remained cheaper than diesel-only generation (\$0.30–0.40/kWh).

These discoveries suggest that while macroeconomic volatility affects absolute cost metrics, hybrid systems maintain a structural cost advantage over fossil-dominant alternatives.

To test resilience under compound macroeconomic instability, three combined scenarios were evaluated:

To enable operational transparency and predictive maintenance, the proposed microgrid architecture incorporates smart prepaid meters with GPRS connectivity, IoT-based inverter telemetry, battery State-of-Health (SoH) sensors, and a cloud-based supervisory control platform. The Pay-As-You-Go (PAYG) model further enhances functionality by enabling mobile money integration, real-time billing, load-limiting features, and remote disconnection for non-payment. Together, this digital architecture minimises revenue leakage, strengthens system reliability, and supports long-term sustainability, addressing key failure points that have historically challenged rural microgrids across Africa.

4.7 Socio-Economic Impact Assessment, Battery Technology Suitability for Humid Tropical Climates

The proposed system supports: 140 households, 1 health centre, 1 primary school, and 17 small commercial enterprises.

Based on productive-use demand estimates:

- i. Welding and sawmill activities increase local income potential by approximately 18–25%.
- ii. Cold storage reduces post-harvest losses by an estimated 15%.
- iii. Reliable power supply to the health centre enables vaccine refrigeration and extended service hours.

Using a conservative multiplier of 0.45 jobs per 10 kW of installed capacity (IEA rural benchmarks), the 1.58 MW renewable installation may indirectly support 6–8 long-term technical and operational jobs within the community.

Battery technologies evaluated are tabulated in Table 8:

Table 7. Multidimensional sensitivity analysis

Scenario	Diesel price	Battery cost	Exchange rate	Discount rate
Base Case	\$1.20/L	\$350/kWh	Baseline	8%
Scenario 1	+30%	–20%	–25% Naira	12%
Scenario 2	+50%	–30%	–35% Naira	15%
Scenario 3	+70%	No cost drop	–40% Naira	18%

Table 8. Comparative battery technology assessment

Technology	Efficiency	Lifespan	Suitability (Tropics)
Lead-acid	75–80%	5–7 yrs	Poor (high temp degradation)
Li-ion (LFP)	90–95%	10–12 yrs	Good
Sodium-ion	85–90%	Emerging	Promising
Flow batteries	75–85%	15+ yrs	High capex

Lithium Iron Phosphate (LFP) chemistry was chosen for the energy storage system because it offers superior thermal stability, reducing the risk of overheating and enhancing safety. Its lower fire risk compared to other chemistries makes it particularly suitable for rural and community-based applications. Additionally, LFP batteries demonstrate better tolerance to humidity, an important factor in tropical environments, and they deliver a lower lifecycle cost, ensuring long-term affordability and sustainability.

4.8 Proposed Community Microgrid Governance Model

To ensure long-term sustainability, a hybrid ownership model is recommended in which the private developer owns 70% of the generation assets, the community energy cooperative holds a 20% equity share, and the local government provides land and regulatory support, totalling 10%. The cooperative would be responsible for collecting tariffs, monitoring consumption, participating in oversight, and receiving dividend returns. Operation and maintenance would be contracted under a 10-year performance-based agreement to guarantee technical reliability. This model significantly reduces the abandonment risk commonly observed in donor-funded rural microgrids.

4.9 Proposed Tariff Structure, Household Affordability and Willingness to Pay

Using the derived LCOE of \$0.197/kWh, a tiered tariff model is proposed, as tabulated in Table 9.

Revenue modelling indicates full cost recovery within 9–11 years under moderate load growth (5% annually). This blended tariff structure balances affordability and financial sustainability.

A household income estimate was derived from regional rural income data for Akwa Ibom, where average monthly earnings range from \$95 to \$140. Applying the 10% income rule, the affordable energy threshold falls between \$9 and \$14 per month. Under the proposed tariff structure, a lifeline household consuming 50 kWh per month at \$0.15 per kWh would spend \$7.50 monthly, while a standard household using 80 kWh per month at \$0.20 per kWh would pay \$16. This means that approximately 68% of households remain within the affordability threshold, with the lifeline tier ensuring inclusion of the poorest segment. Overall, this analysis suggests that the \$0.197/kWh LCOE can be translated into a socially viable tariff structure through tiered pricing.

Compared with before optimisation, Table 10 provides a broad overview.

Table 9. Proposed tariff framework

Consumer class	Tariff (US\$/kWh)	Rationale
Lifeline (≤ 50 kWh/month)	\$0.15	Subsidised for low-income households
Residential Standard	\$0.20	Cost-reflective
Commercial	\$0.23	Cross-subsidises residential users
Productive Use	\$0.18	Incentivises economic activity

Table 10. Comparison of pre-optimised baseline vs. HOMER-optimised system

Component	Baseline design (section 3)	HOMER-optimised system	Rationale for change
Solar PV	270 kW	393 kW	Prevents significant overcapitalisation; optimised to match load profile and local solar yield.
Wind Turbine	80 kW (4 x 20 kW)	20 kW (1 x 20 kW)	Low wind capacity factor (~25%) limits economic viability; reduced to a single supplementary unit.
Battery Storage	562 kWh	2,515 kWh	Crucial for ensuring reliability without diesel; provides 47 hours of autonomy to manage renewable intermittency.
Diesel Generator	100 kW	0 kW	Eliminated due to high operational costs and emissions; reliability achieved via renewables + storage.
Converter	380 kW	144 kW	Right-sized to handle the peak load of the optimised system, reducing capital cost.

These limitations, however, do not undermine the study's core conclusions; instead, they highlight areas for methodological enhancement in future work.

5. Conclusion

This study presented a detailed optimisation and techno-economic analysis of a solar–wind–diesel hybrid microgrid for sustainable rural electrification in Ibiaku Ikot Oku, Nigeria. The HOMER Pro simulation results demonstrate that the PV–Wind–Diesel–Battery configuration yields the most cost-effective and environmentally favourable solution among the modelled alternatives. The optimised system achieved a Net Present Cost (NPC) of \$1.06 million, a Levelized Cost of Energy (LCOE) of \$0.197/kWh, and a renewable fraction of 78.2%, representing a significant improvement over a conventional diesel-only system. Furthermore, the system reduced CO₂ emissions by 45.7%, demonstrating its environmental superiority and alignment with Nigeria's climate commitments under the Paris Agreement (Nigeria, 2025). These outcomes indicate that hybrid microgrids can provide reliable, affordable, and clean electricity to rural communities in Nigeria, while reducing their dependence on imported fossil fuels. The integration of complementary renewable resources (solar and wind) enhances system reliability, minimises fuel cost exposure, and improves lifecycle economics. The findings also reinforce earlier conclusions that multi-source hybridisation represents a pragmatic and scalable solution to rural electrification challenges in sub-Saharan Africa. Additionally, this work provides transparent cost breakdowns, operational insights, and sensitivity analyses that can guide policymakers, investors, and energy developers in planning and implementing microgrids.

6. Policy Implications, Study Limitations and Future Work

6.1 Policy Implications

From a policy perspective, the study supports the decentralisation objectives of Nigeria's Electricity Act (2023), which empowers state-level and private sector participation in distributed generation. The derived LCOE benchmark of \$0.197/kWh can inform tariff design and financial modelling for mini-grid developers under the NEP and Rural Electrification Fund (REF) schemes. Furthermore, the achieved emission reduction potential positions hybrid microgrids as essential components in achieving Nigeria's 30% renewable energy target by 2030, as articulated in the NREEEP and NDC 2024 commitments (Nigeria, 2025).

6.2 Limitations of the Study

While the results obtained are robust and contextually relevant, specific limitations exist:

- i. **Modelling simplifications:** The HOMER Pro platform assumes ideal control strategies and may not capture dynamic operational constraints, such as inverter efficiency degradation or transient faults.
- ii. **Static resource assumptions:** The use of annual-average irradiance and wind speed data limits the representation of temporal variability.
- iii. **Economic assumptions:** A fixed discount rate and constant fuel escalation rates may not accurately reflect the macroeconomic volatility in Nigeria's energy market.

6.3 Future Research Directions

To deepen understanding and strengthen the applicability of hybrid microgrid solutions, future work should consider the following:

- i. Incorporation of advanced optimisation algorithms such as Particle Swarm Optimisation (PSO), Genetic Algorithms (GA), or Reinforcement Learning-based controllers for improved component sizing and control precision.
- ii. Dynamic and probabilistic modelling using Monte Carlo simulations or time-series forecasting to account for meteorological and load uncertainties.
- iii. Integration of emerging technologies, including hydrogen storage systems and supercapacitors, to enhance reliability and system flexibility.
- iv. Field validation and socio-economic analysis, involving real-time performance monitoring and community-level impact assessment, to verify model predictions and inform adaptive policies.
- v. Integration of smart grid and IoT technologies for remote monitoring, predictive maintenance, and adaptive energy management, enabling data-driven decision-making in microgrid operations.

Disclaimer (Artificial Intelligence)

Author(s) hereby declare that NO generative AI technologies such as Large Language Models (ChatGPT, COPILOT, etc.) and text-to-image generators have been used during the writing or editing of this manuscript.

Competing Interests

Authors have declared that they have no known competing financial interests OR non-financial interests OR personal relationships that could have appeared to influence the work reported in this paper.

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